25900405 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCFSC201 - FORENSIC CHEMISTRY

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions

Answer any **10** questions

Each question carries **2** marks

1. What is chain of custody? [K] / [CO1]
2. Explain the principle of the chromotropic acid test for detecting methyl alcohol [U] / [CO2]
3. Why is it important to find the point of origin in fire investigation? [A] / [CO3]
4. What does the Rf value indicate in TLC analysis of petroleum products? [An] / [CO4]
5. Write two safety Precautions in forensic chemical lab. [K] / [CO1]
6. What is a non-alcoholic beverage? [U] / [CO2]
7. Suggest one unique feature of fats that could aid in crime reconstruction. [An] / [CO2]
8. How can UV spectrophotometry help in ink analysis? [A] / [CO3]
9. What is the principle of Thin Layer Chromatography (TLC)? [A] / [CO4]
10. Describe the role of congeners in alcoholic beverages. [K] / [CO2]

11. Why is it important to compare lubricating oils at both physical and chemical levels? [An] / [CO1]

12. Explain the role of phenolphthalein in detecting trap cases [K] / [CO2]

[2x10 = 20]

Part B

Short Essay Type Questions

Answer any **5** questions

Each question carries **4** marks

13. What is the Rhodamine test? How is it used in dye examination? [K] / [CO1]

14. Describe how solid-phase extraction can be applied for drug evidence purification. [An] / [CO2]

15. Reflect on the importance of different petroleum fractions in daily life and industry. [A] / [CO4]

16. Compare the forensic value of oils and fats when recovered from a suspect's clothing. [U] / [CO1]

17. Explain the difference between Indian Made Foreign Liquor (IMFL) and non-alcoholic beverages. [K] / [CO2]

18. Define the main types of petroleum products and give one example of each [A] / [CO4]

19. Examine the differences between light produced by a flame and other light sources in forensic investigation. [A] / [CO4]

[4x5 = 20]

Part C

Essay Type Questions

Answer any **1** questions

Each question carries **10** marks

20. Analyse the role of cation and anion identification in the forensic detection of explosives, using examples to illustrate how specific ions point to different explosive materials. [An] / [CO2]

21. Explain the differences between country-made liquor, illicit liquor, denatured spirits, Indian Made Foreign Liquor (IMFL), and non-alcoholic beverages, highlighting their key characteristics and legal status. [A] / [CO2]

[10x1 = 10]